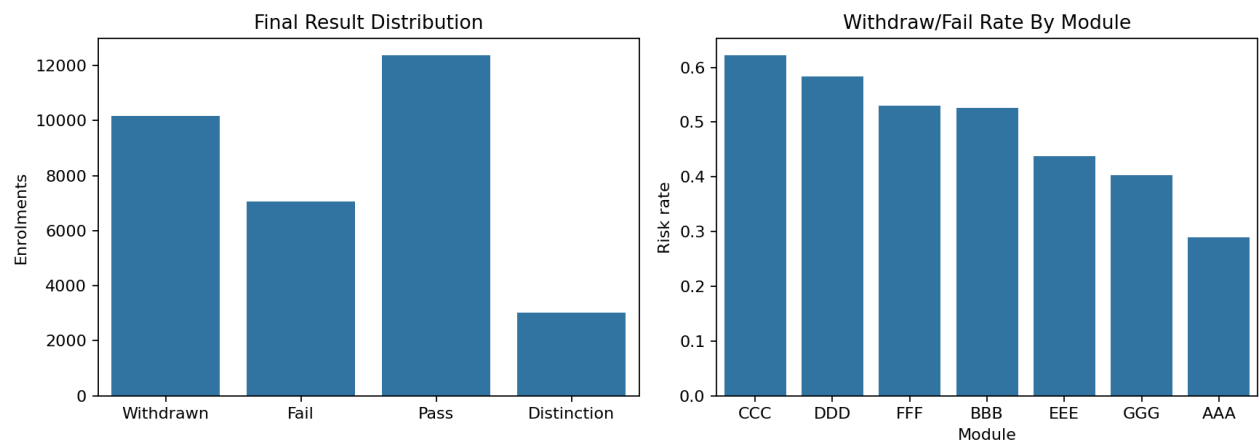


Studor PathAI: Baseline Product Prototype

This analysis converts OULAD behavioral, assessment, and student context data into three product-facing assets: a weekly engagement score, a Week 6 disengagement risk model, and next-course recommendations.

The Week 6 model predicts Withdrawn/Fail versus Pass/Distinction. The urgent tier now alerts 57.7% of the test cohort at precision 0.76 and recall 0.83; the broader watchlist alerts 85.0% for monitoring.



Data Cleaning And Leakage Controls

The enrolment key is student + module + presentation; student ID alone is not unique because learners can take multiple modules. Cleaning is conservative: semantic issues are audited rather than silently rewriting the official target label. Negative VLE dates are treated as valid pre-start engagement before day 0.

Check	Issue Count	Policy
studentInfo key uniqueness	0	Use student-module-presentation as the enrolment key.
studentRegistration key uniqueness	0	Use student-module-presentation as the enrolment key.
Unregistered enrolments not labelled Withdrawn	9	Audit only; official final_result remains the supervised target.
Withdrawn enrolments without unregistration date	93	Do not use date_unregistration as a predictive feature because it can occur after Week 6.
Missing score rows	173	Keep missing scores as missing; median-impute only inside model preprocessing.
Invalid score rows outside 0-100	0	Invalid values would be excluded from score aggregates.
Assessment rows with missing due date	11	Exclude missing-date assessments from weekly due/submission features.
Negative VLE activity dates (pre-start engagement)	688988	Valid pre-presentation activity; date is days before module start. Used as proactivity signal.
VLE activity after module presentation length	0	Exclude from weekly engagement panel.
Missing VLE week_from/week_to metadata	10486	Do not use week_from/week_to as model features.

Feature	Available Week 6?	Used?	Reason
Pre-start VLE activity (negative dates)	Yes	Yes	Observed before day 0; reflects eagerness to begin course materials.
Week 1-6 VLE clicks	Yes	Yes	Observed before the Week 6 prediction point.
Week 1-6 active days	Yes	Yes	Observed before the Week 6 prediction point.
Week 1-6 activity diversity	Yes	Yes	Observed before the Week 6 prediction point.
Early non-exam assessment submissions	Yes	Yes	Only due/submitted work by Week 6 is used.
Student demographics/context	Yes	Yes	Known at registration or course enrolment.
Engagement score	Yes	Yes	Computed from Week 1-6 behavior using train-derived score weights.

Task 1: Dynamic Engagement Score

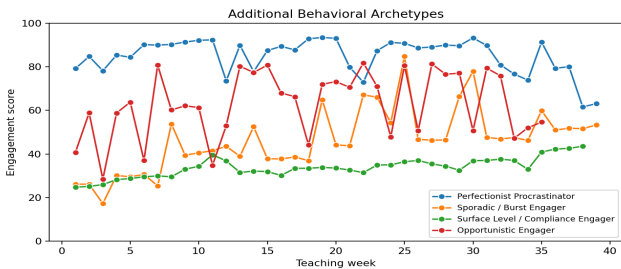
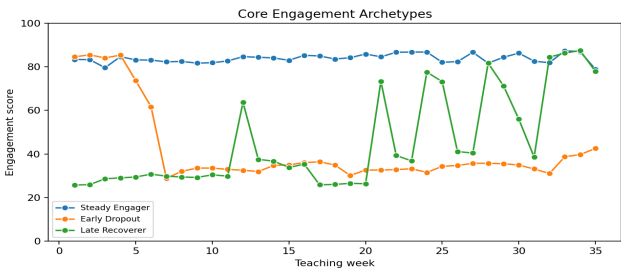
The 0-100 engagement score updates weekly and is peer-normalized within module, presentation, and week. Component weights are derived from the training split by measuring how well each component separates successful students from withdraw/fail students at Week 6, then rounded into a readable scorecard.

Most data-backed Week 6 signals: engagement_score $r=-0.516$; assessment_submitted_ratio $r=-0.486$; zero_click_weeks_cum $r=+0.473$; weekly_clicks_norm $r=-0.468$; missed_assessments_cum $r=+0.453$.

Component	Success AUC	Weight
Peer-normalized click volume	0.774	15%
Module-material study frequency	0.741	15%
Activity diversity	0.762	15%
Recent activity	0.773	15%
Pre-start course material engagement	0.690	10%
Active-day consistency	0.778	10%
Assessment completion	0.718	10%
Low-weight assignment completion	0.684	10%
Three-week trend	0.481	0%
Assessment punctuality	0.469	0%

Scorecard guardrail: components with train success AUC ≤ 0.50 are assigned 0% weight because they are weaker than random as score components in the current definition. They can remain exploratory features, but they should not move the trusted 0-100 score.

Archetype	Feature signature
Steady Engager	High mean engagement score, low score volatility, strong active-day consistency.
Early Dropout	Sharp negative early-to-late score delta, low late score, often withdrawn final result.
Late Recoverer	Large positive score delta from early weeks to late weeks.
Sporadic / Burst Engager	High score range, high max score, and several zero-click weeks.
Perfectionist Procrastinator	Successful/high-scoring enrolment with late submissions and lower punctuality.
Surface Level / Compliance Engager	High assessment completion but low activity diversity and low material-study frequency.
Opportunistic Engager	High material-click share and engagement spikes, but weaker active-week consistency.

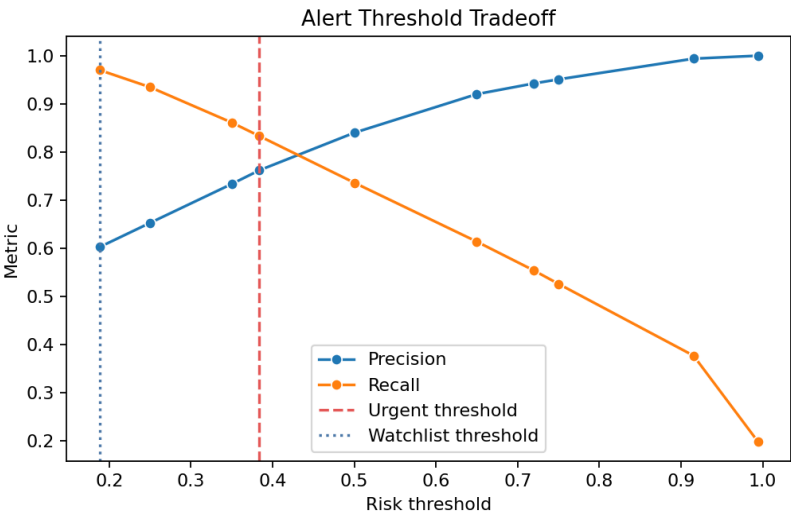
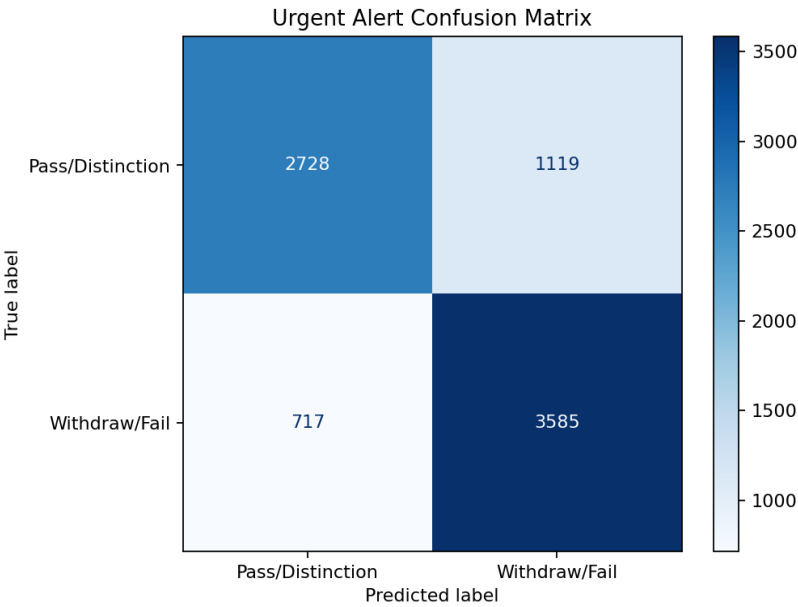


Task 2: Week 6 Risk Model

The supervised model uses only features available through Week 6. It includes the Task 1 engagement score, a Week 6-safe behavioural archetype, transformed profile context (education order, IMD midpoint, age order, normalized credit load), resource-mix ratios, and selected interaction features. Candidate models are compared with cross-validated F1, PR-AUC, and ROC-AUC for the urgent alert, while a high-recall F2 threshold remains available as a watchlist.

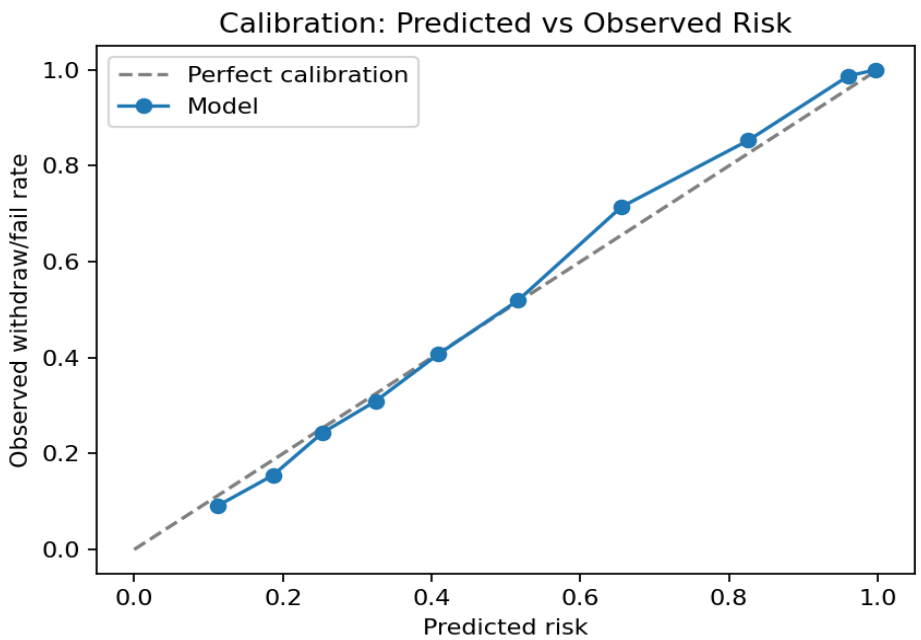
Urgent threshold 0.38: precision 0.762, recall 0.833, F1 0.796, ROC-AUC 0.872. Confusion matrix counts: TN=2728, FP=1119, FN=717, TP=3585.

Tier	Threshold	Precision	Recall	F1	Alerts	Alert Share	FP	FN
Urgent	0.38	0.76	0.83	0.80	4704	57.7%	1119	717
Watchlist	0.19	0.60	0.97	0.74	6924	85.0%	2750	128
Capacity 20%	0.92	0.99	0.38	0.55	1634	20.1%	10	2678
Fixed 0.50	0.50	0.84	0.74	0.78	3771	46.3%	603	1134



Calibration And Advisor Alert Design

The alert design uses two tiers. Urgent alerts use a balanced F1 threshold to create advisor tasks with better precision; watchlist alerts use a higher-recall F2 threshold for monitoring and automated nudges.



Group	Feature	Mechanism
Overall	avg_score_so_far_6	Early academic struggle may signal content difficulty before final failure.
Overall	first_click_day_6	High-ranking driver in permutation importance.
Overall	module_code	High-ranking driver in permutation importance.
Behavioral	engagement_score_6	Combines recency, consistency, diversity, and assessment behavior into a trajectory signal.
Behavioral	assessment_submitted_ratio_6	Missing early assessments is both predictive and directly actionable for advisors.
Behavioral	weekly_clicks_norm_6	Behavioral signal advisors can investigate or act on.

PathAI Advisor Alert	
Student	S-10482
Course	DDD-2014J
Risk tier	Urgent
Predicted risk	78%
Why flagged	No VLE activity in 12 days; engagement score dropped from 61 to 34 over two weeks; first assessment due by Week 6 was not submitted.
Suggested action	Send check-in within 48 hours. Ask about workload, access issues, and confidence with course material. Offer academic support session.

Task 3: Course Recommendations

The recommender primarily serves students planning a next semester, while advisors can use the same output in guidance conversations. Content-based recommendations use education, age band, historical module success, and prior engagement band. Collaborative filtering uses cosine similarity over successful module patterns.

Metric	Value
Holdout students	2479
Content hit@3	0.111
Collaborative hit@3	0.594
Content coverage	5/7
Collaborative coverage	7/7
Cold start	AAA, EEE, GGG

Limitations: OULAD has a small module catalog, recommendation evaluation is a proxy holdout, and production deployment would need advisor feedback loops plus monitoring by module and presentation.

90-day roadmap	1. Pilot tiered alerts with advisors and track accepted interventions.
	2. Add advisor feedback labels to calibrate thresholds by module workload.
	3. Extend recommendations with richer course metadata and student goals.